

PILOT – Cross-Domain Invariants (Identity, Privilege, Endpoint)

Objective:

Define deterministic constraints across core IT domains that must hold for system activity to be valid.

1. Identity Domain Invariants

1. Authenticated Identity Requirement

Any action attributed to a user identity must be associated with a valid authenticated session at the time of execution.

2. Identity Existence Consistency

Any active identity used must exist as a valid identity object in the system of record at that time.

3. Role Assignment Integrity

Any action requiring a role must be consistent with the user's role assignment at that exact time.

4. Group Membership Consistency

If access depends on group membership, that membership must be valid at the time of access.

5. Identity Lifecycle Consistency

Disabled, suspended, deleted, or expired identities must not perform new authenticated actions.

6. Session–Identity Binding

Any active session must map to a valid identity and must not exist independently.

7. Authentication Method Consistency

Where policy requires specific authentication conditions, they must be satisfied.

8. Temporal Consistency of Identity State

Authentication, role state, and downstream access must align in time.

9. Identity Uniqueness / Non-Ambiguity

A single action must not resolve to conflicting identities.

10. Cross-Domain Identity Coherence

Identity state must remain consistent with privilege and endpoint state.

2. Privilege Domain Invariants

1. Authenticated Session Requirement

Any privileged action must be associated with a valid, active authenticated session.

2. Identity–Privilege Consistency

Privileged action must match identity state and role assignment.

3. Credential–Session Binding

Credential use must map to a valid session.

4. Endpoint Context Consistency

Privileged action must originate from a valid endpoint.

5. Process–Identity Alignment

Processes performing privileged actions must align with identity context.

6. Temporal Consistency

Authentication, credential use, and execution must align in time.

7. Access Path Validity

Identity → credential → access path must be continuous.

8. Session Continuity

Privileged activity must occur within a continuous session.

9. Role Transition Integrity

Privilege elevation must correspond to valid role change.

10. Cross-Domain Coherence

Identity, privilege, and endpoint states must be consistent.

3. Endpoint Domain Invariants

1. Endpoint Existence Requirement

Actions must originate from valid, known endpoints.

2. User–Endpoint Binding

Actions must map to a valid user or service context on that endpoint.

3. Process Lineage Integrity

Processes must have valid execution lineage.

4. Execution Context Consistency

Actions must align with user and host execution context.

5. Endpoint State Validity

Endpoint must be in valid operational state.

6. Session–Process Consistency

Processes must align with active sessions.

7. Privilege–Execution Consistency

Elevated execution must match authorised privilege context.

8. Temporal Consistency of Execution

Login, process launch, and execution must align in time.

9. Asset Ownership Consistency

Actions must originate from correctly assigned endpoints.

10. Cross-Domain Endpoint Coherence

Endpoint state must align with identity and privilege state.

Bottom Line

These invariants define what must be true for activity to be possible across identity, privilege, and endpoint domains.

Violations indicate causally inconsistent system states, not merely suspicious behaviour.